

PSimPy - Predictive and probabilistic simulation with Python



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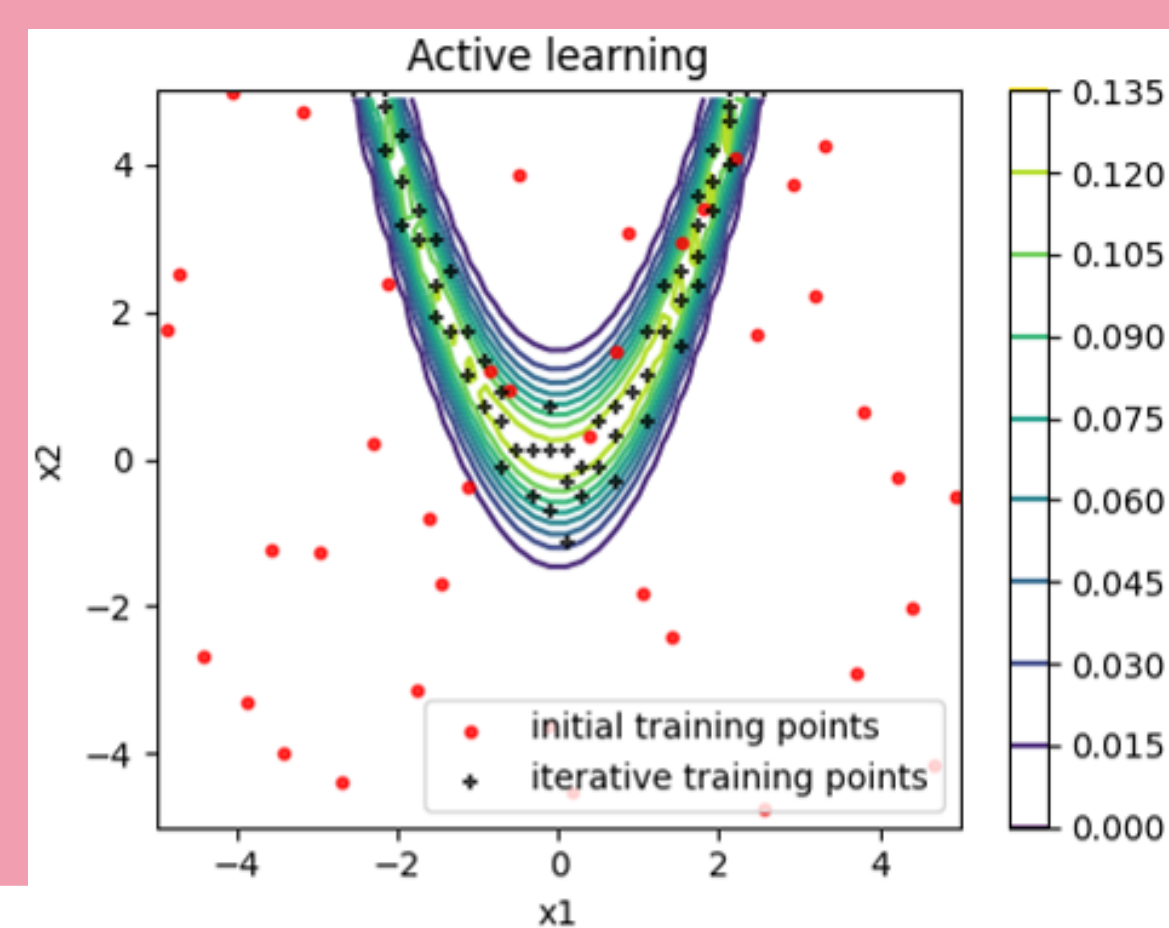
Gaussian Process, Multiphysics Simulation

Summary

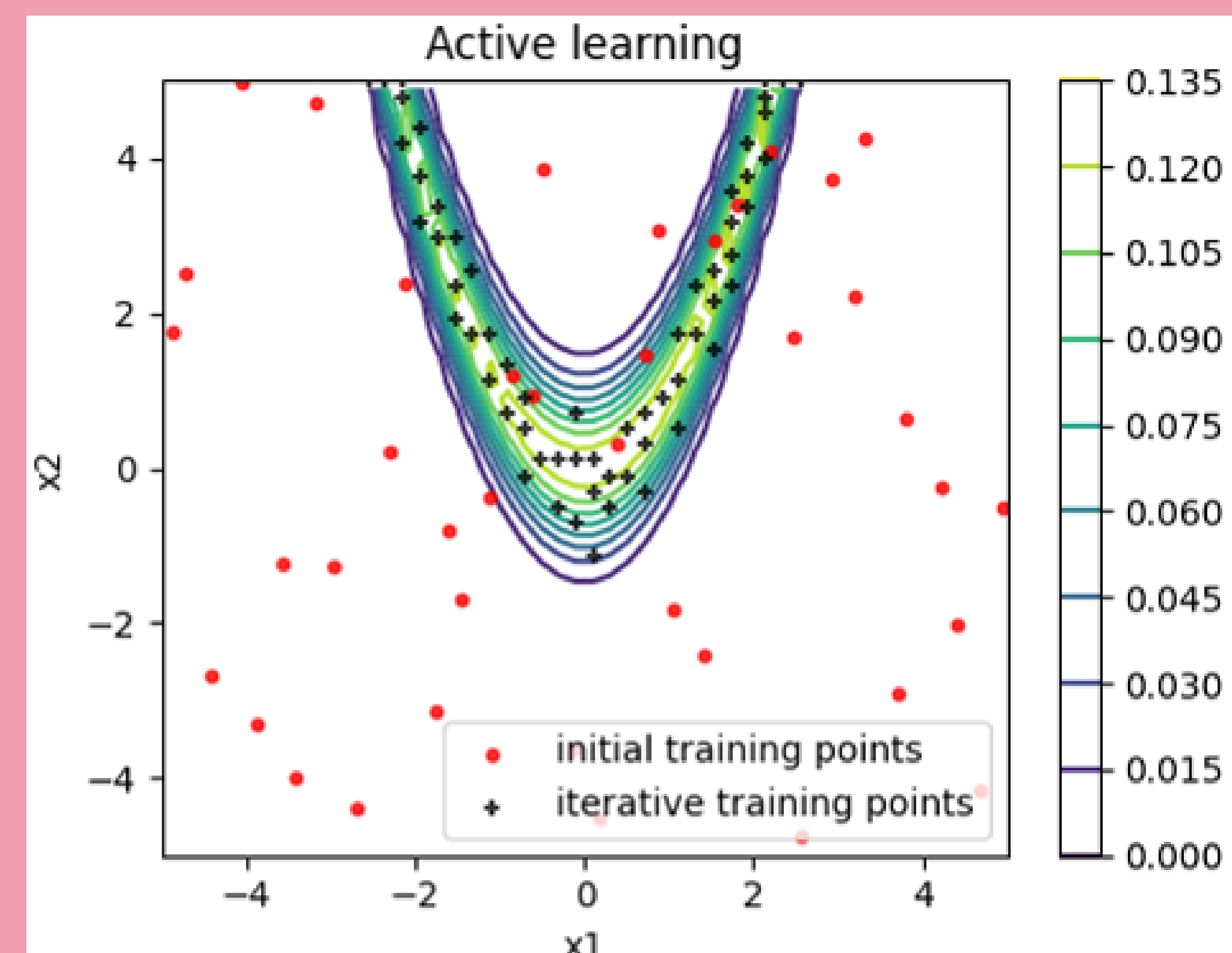
Train an GP Emulator and save compute time!

	training	evaluation	2500 evals
simulator	-	32 min	56 days
emulator	4.5 days	0.05 sec	2 min

- Python wrapper
- Robust GP Emulation
- Lightweight Substitute



Software in Action



Identify key drivers. Calibrate with data.

- Run sensitivity analysis using Sobol indices
- Sample efficiently using Saltelli, LHS, or Metropolis Hastings

Want to build unified,
robust and sustainable
UQ workflows?

PSimPy

Bibliography

- Set up a Bayesian problem
- Kick off with a small batch of simulations and a GP Emulator
- Active learning automatically picks the next input points to improve calibration
- Estimate the posterior from the emulator and visualize it

Active-learning calibration

Example

Bibliography